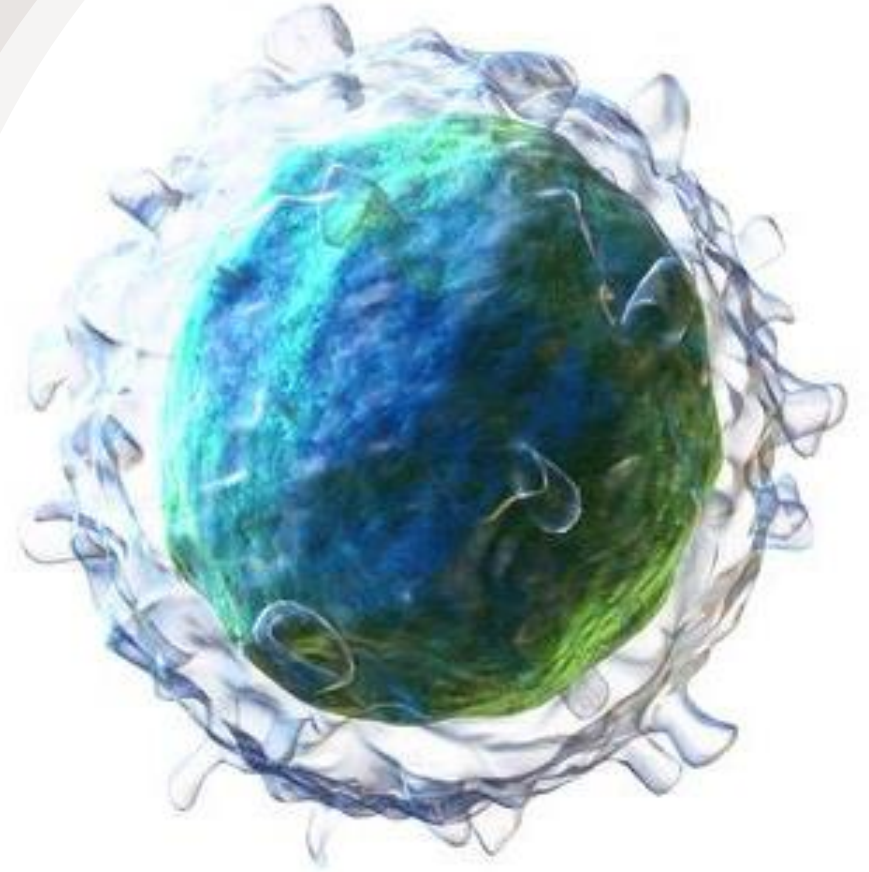


Isolation of B Lymphocytes



Lymphocyte

B cell



- **Lymphoid organs** : sites where Lymphocytes developing, Diversity and maturation of lymphocytes occur.

- Two types of Lymphoid organs:

- **Primary Lymphoid organs** : where the lymphocytes development and maturation occur include bone marrow and Thymus

- **Secondary lymphoid organs** : where the immune response occur, include spleen and lymph nodes and others.

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- Spleen : largest secondary lymphoid organs, and the primary site of Antibodies production, contain billions of mature B cells.
 - We isolate Antibodies producing cells from secondary lymphoid organs not primary lymphoid organs why?
 - Because the cells present in primary lymphoid organs are Immature, which mean cant secrete Antibodies, while the B cells present in Spleen are mature.
 - The best organ for B cells isolation is spleen, because it rich source of B cells.

Spleen cells

- There are several commonly used methods for preparing single cell suspension from spleen.

A. Pressing tissue through a fine mesh strainer

b. Dispersion of cells by fine jets of media by different needle sizes

c. Dispersion of cells between 2 frosted slides

One mouse spleen yields about 5×10^7 - 2×10^8 cells.

Frosted slides



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- Cell strainer: mesh like structure, sterile, rapid and easy use device for isolation primary cells to obtain uniform single cell suspension from tissues.
 - Cell strainer made from strong nylon, with three sizes :
 - 40 micro meter
 - 70 micro meter
 - 100 micro meter

CELL STRAINERS



Procedure

- After sacrificed the mouse using Alcohol for Sterilization (before dissection)
- Place the spleen into the cell strainer. Using the plunger end of the syringe, mash the spleen through the cell strainer into the petri dish.
- Rinse the cell with 5mL DMEM-10. Discard the strainer.
- Transfer the suspended cells to a 15mL conical.
- Spin cells at 800xg for 3 minutes.
- Discard supernatant and resuspend pellet in 1mL ammonium chloride lysis buffer. Incubate at RT for 5-10 minutes. Add 9mL DMEM-10 and spin as before.
- Discard supernatant and resuspend pellet in 3mL DMEM-10, discarding any dead cell mass.

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- Ammonium chloride lysis buffer normally the number of RBCs is much greater than WBC, Ammonium chloride lysis buffer lysing the RBCs and Platelets in short times, also WBCs, but WBC require long time to lysed.
 - After RBCs and platelets lysed, we wash the Lysis buffer.